

Three Preconditions for Latent Opponent Modeling in Adversarial Co-Training: An Empirical Study in a Resource-Augmented Predator–Prey Game

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Abstract

We are building toward an adversarial co-training method in which one team (*blue*) plans against a learned, uncertainty-aware model of its opponent (*red*): a variational encoder maps a red trajectory to a latent strategy variable z , a latent-conditioned policy reconstructs red’s behavior, and a model-based planner samples that policy at each search node. The method rests on three empirical preconditions that are usually *assumed*: (P1) the opponent exhibits genuinely distinct strategies; (P2) those strategies are recoverable by an unsupervised encoder into a latent that separates them; and (P3) a strong co-trained best response is actually brittle to an out-of-distribution opponent, so that opponent-aware planning has something to fix. We test all three in a resource-augmented predator–prey game (**SimpleTagResourcesMPE**, JaxMARL). A prior experiment found that a *dominated* prey expresses its resource-collection strategy only as a weak aggregate perturbation, blocking all three. Restoring a balanced regime—specialist prey co-trained on a single resource layout, with predator speed reduced so the prey can execute a route—makes all three preconditions hold: the circle- and corners-specialist prey trace a diamond and a square respectively, separable from occupancy at 0.98 (chance 0.50); an unsupervised VAE on the trajectory occupancy histogram recovers the strategy with adjusted Rand index 0.87 and a latent linear-probe accuracy of 0.99 given ≥ 100 steps of observation; and MAPPO predators co-trained against one prey suffer a 43% drop in capture rate when faced with the unseen prey. Building on these, we realize the proposal’s Part 1: a single behavior-cloned policy $\pi_\psi(a \mid s, z)$ that reproduces either specialist by swapping its latent, whose route is controllable by z at 0.85 even with the environment held fixed. We give the formal setup, the encoder and cloning objectives, and the metric definitions, and discuss the regime-dependence of each result.

1 Introduction

In adversarial multi-agent reinforcement learning (MARL), each team faces a non-stationary environment because the opposing team is itself learning. A recurring failure mode is *overfitting to the opponent*: a team learns a best response to its current adversary and loses robustness to strategies it has not recently seen. The method this paper supports addresses that failure by giving the planning team an explicit, uncertainty-aware model of its opponent. Concretely (Parts 1–2 of the proposal):

Part 1 (encoder).

A variational encoder $q_\phi(z \mid \tau^{\text{red}})$ maps a red (opponent) trajectory τ^{red} to a latent strategy

variable z , trained by maximizing an evidence lower bound (ELBO). A latent-conditioned policy $\pi_\psi^{\text{red}}(a \mid s, z)$ then reconstructs red’s behavior, by behavior cloning or contrastive preference learning.

Part 2 (planner).

A model-based (MuZero-style) planner expands a search tree in which blue actions branch and red actions are *sampled* from $\pi_\psi^{\text{red}}(\cdot \mid s, z)$, with a latent-conditioned value function $v^{\text{blue}}(s, z)$ scoring nodes.

This machinery is only worth building if three things are true of the domain. The opponent must *have* distinct strategies to model (P1); those strategies must be *encodable* into z by an unsupervised objective (P2); and the co-trained policy we intend to improve must in fact be *brittle* to out-of-distribution opponents (P3), or there is nothing for opponent-aware planning to fix. This paper isolates and tests P1–P3 directly, before investing in the encoder–planner stack.

Prior negative result. An earlier experiment built a resource-augmented predator–prey task expressly to make strategy separable, then found the separation absent: a single prey trained on a *mixed* circle/corners placement produced trajectories whose layout was not recoverable even by a supervised probe (accuracy $\approx$ chance), and an unsupervised trajectory VAE reached adjusted Rand index ARI ≈ 0.008 against the placement label—essentially the same null as the earlier reward-shaping study. We trace this to opponent *domination*: three coordinated predators force the prey into near-pure evasion across the whole arena, so the resource layout perturbs its occupancy only weakly and never on a single trajectory. The contribution here is to show that, in a balanced regime where the prey expresses its strategy, all three preconditions hold, with the formal setup and metrics needed to state them precisely.

2 Problem setup

2.1 Markov game

We model the task as a partially observed Markov game $\langle \mathcal{N}, \mathcal{S}, \{\mathcal{A}_i\}, \{\mathcal{O}_i\}, P, \{R_i\}, \gamma \rangle$ with agents $\mathcal{N} = P \cup \{q\}$ split into a predator team $P = \{1, 2, 3\}$ and a single prey q . At step t , agent i receives an observation $o_i^t \in \mathcal{O}_i$, selects a discrete action $a_i^t \in \mathcal{A}_i = \{\text{stay}, \leftarrow, \rightarrow, \downarrow, \uparrow\}$, and the joint action drives the MPE point-mass dynamics $s^{t+1} \sim P(\cdot \mid s^t, a^t)$. Episodes last $H = 25$ steps. Predators share weights within the team; the two teams are trained as separate actor–critic (MAPPO) or Q -learning (IQL) pairs, so the game is adversarial, not cooperative.

2.2 Rewards

Let $\mathbf{1}[\cdot]$ be the indicator and write $c_i^t = \mathbf{1}[\|x_i^t - x_q^t\| < \rho_i + \rho_q]$ for the event that predator i is in contact with the prey at step t , where x are positions and ρ are body radii. The predator team receives a *shared* reward proportional to the number of predators in contact,

$$r^{\text{pred},t} = \lambda_{\text{tag}} \sum_{i \in P} c_i^t, \quad \lambda_{\text{tag}} = 10, \quad (1)$$

and every predator receives this same value. The prey receives the negative tag reward, a boundary penalty $b(x_q^t) \geq 0$ discouraging it from leaving the arena, and a collection bonus:

$$r^{\text{prey},t} = -\lambda_{\text{tag}} \sum_{i \in P} c_i^t - b(x_q^t) + \lambda_{\text{col}} \sum_{j=1}^4 \Delta_j^t, \quad (2)$$

where $\Delta_j^t \in \{0,1\}$ marks resource j being collected for the first time at step t . A resource is collected when the prey approaches within $r_{\text{col}} = 0.15$; only the prey observes resource positions and only the prey can collect. We define *captures per episode* as the episode tag count recovered from the shared predator return,

$$\text{cap} = \frac{1}{\lambda_{\text{tag}}} \sum_{t=1}^H r^{\text{pred},t}. \quad (3)$$

2.3 Resource layouts

Four resources are placed at reset according to a layout ℓ :

$$\text{circle: } g_j = 0.6 (\cos \theta_j, \sin \theta_j), \theta_j = \frac{\pi}{2}j; \quad \text{corners: } g_j \in \{\pm 0.8\}^2. \quad (4)$$

The two layouts induce different optimal collection routes—a loop around the ring versus a tour of the four corners—which is the strategy signal P1–P3 are about. Two fixed obstacles sit at $(\pm 0.5, \pm 0.5)$.

2.4 The encoder we are building toward

Part 1 learns $q_\phi(z \mid \tau^{\text{red}})$ and a decoder $p_\theta(\tau^{\text{red}} \mid z)$ by maximizing the ELBO

$$\mathcal{L}(\phi, \theta) = \underbrace{\mathbb{E}_{q_\phi(z \mid \tau^{\text{red}})} [\log p_\theta(\tau^{\text{red}} \mid z)]}_{\text{reconstruction}} - \underbrace{D_{\text{KL}}(q_\phi(z \mid \tau^{\text{red}}) \parallel p(z))}_{\text{regularizer}}, \quad p(z) = \mathcal{N}(0, I). \quad (5)$$

P2 below is exactly the question of whether the latent learned by Eq. (5) separates red’s strategies. A key empirical choice, motivated by the prior negative result, is the encoder *input*. A raw position sequence is dominated by evasion and start-position noise; the layout signal lives in *where* the prey spends time. We therefore featurize a trajectory $\tau = (x^1, \dots, x^L)$ by its normalized 2D occupancy histogram over a $B \times B$ grid on $[-1.6, 1.6]^2$,

$$\Phi_B(\tau)_{kl} = \frac{1}{L} \sum_{t=1}^L \mathbf{1}[x^t \in \text{bin}_{kl}] \in \Delta^{B^2-1}, \quad B = 12, \quad (6)$$

and use $\Phi_B(\tau)$ as the input to the encoder of Eq. (5).

3 The three preconditions as measurable hypotheses

Metrics. For a feature map φ and binary placement labels $y \in \{0,1\}$ (circle vs corners, recovered from resource geometry and never shown to any encoder), the *linear-probe accuracy* $\text{Acc}(\varphi)$ is the 5-fold cross-validated accuracy of ℓ_2 -regularized logistic regression predicting y from φ . For an

unsupervised clustering $\hat{y}$ (here a two-component Gaussian mixture fit to the latent z), the *adjusted Rand index* $\text{ARI}(\hat{y}, y) \in [-1, 1]$ measures agreement with the true labels corrected for chance (0 at chance, 1 for a perfect match). We state each precondition as a hypothesis with its estimator:

P1 (distinct strategies).

$\text{Acc}(\Phi_B(\tau)) \gg 0.5$: the placement is linearly decodable from a trajectory’s occupancy, i.e. the two specialists occupy the arena differently.

P2 (encodable).

For $z \sim q_\phi(\cdot \mid \Phi_B(\tau))$ trained *unsupervised* by Eq. (5), both $\text{Acc}(z) \gg 0.5$ (the latent retains strategy) and $\text{ARI} \gg 0$ (an unsupervised reading of the latent recovers it).

P3 (brittleness).

Let $M_{\ell\ell'} = \mathbb{E}[\text{cap}]$ when a predator co-trained on layout ℓ faces a prey co-trained on layout ℓ' , each in its native layout. Then the in-distribution (diagonal) mean exceeds the out-of-distribution (off-diagonal) mean: $\bar{M}_{\text{in}} = \frac{1}{2}(M_{cc} + M_{kk}) > \bar{M}_{\text{ood}} = \frac{1}{2}(M_{ck} + M_{kc})$. We report the relative brittleness $\beta = 1 - \bar{M}_{\text{ood}}/\bar{M}_{\text{in}}$.

3.1 Experimental regime

To make red express strategy we (i) train *specialists*: a separate predator+prey co-training run on each fixed layout (circle-only, corners-only), three seeds each; and (ii) balance the game so the prey is not dominated, by reducing predator top speed $1.0 \rightarrow 0.6$ and acceleration $3.0 \rightarrow 1.5$ (prey unchanged at 1.3/4.0) and raising the collection bonus $\lambda_{\text{col}} : 5 \rightarrow 10$. All other settings match the base task. IQL/OA-IQL use a 128-unit MLP, replay buffer 5000, batch 32, $\epsilon : 1.0 \rightarrow 0.05$, $\gamma = 0.9$, learning rate 0.005. MAPPO uses a per-team actor with a centralized critic over the concatenated team observation, 32 environments $\times$ 128 steps, 4 PPO epochs, $\gamma = 0.99$, $\lambda_{\text{GAE}} = 0.95$. Evaluation uses 200 greedy episodes per cell for P3 and 300 trajectories per specialist for P1/P2.

4 Results

4.1 P1 — different layouts produce different strategies

The two specialists trace visibly different routes (Figure 1): the circle-prey loops a diamond between the four ring resources, the corners-prey patrols a square around the perimeter. Pooling each trajectory into its occupancy histogram (Eq. 6), placement is linearly decodable at $\text{Acc} = 0.98$ for the IQL prey and 0.96 for MAPPO, against a chance rate of 0.50 (Table 1). P1 holds.

Table 1: **P1.** Linear-probe placement accuracy from a single trajectory’s occupancy histogram (chance = 0.50).

Prey policy	Occupancy-probe accuracy	Chance
IQL specialist	0.98	0.50
MAPPO specialist	0.96	0.50
(prior) mixed-placement prey	≈ 0.50	0.50

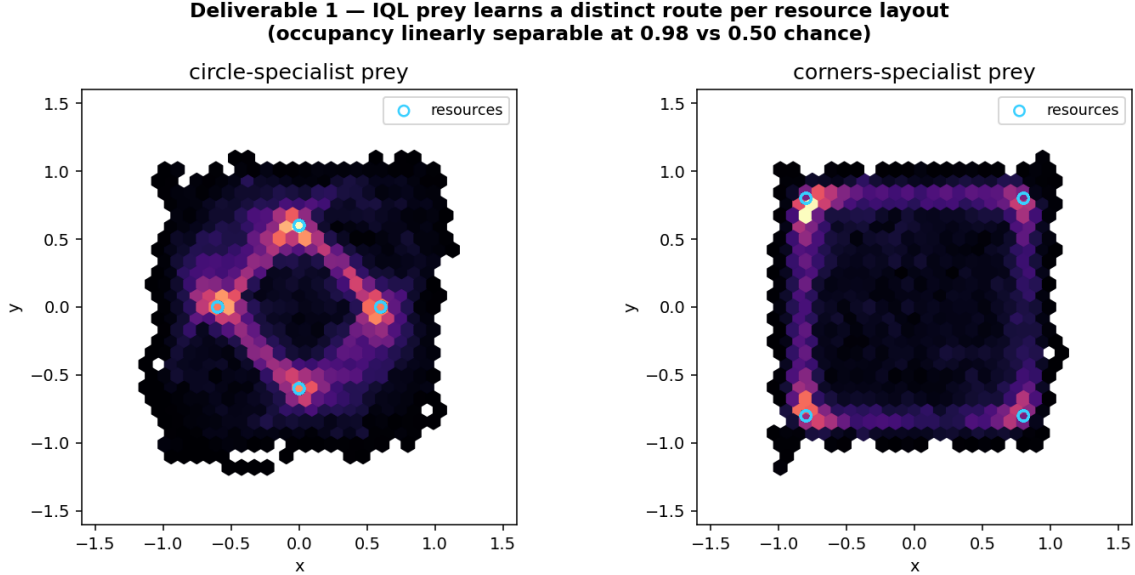


Figure 1: **P1.** Prey occupancy density (pooled over 300 episodes per specialist; cyan rings mark resource positions). The circle-specialist’s route is a diamond between ring resources; the corners-specialist’s is a perimeter square. Occupancy is linearly separable at 0.98.

4.2 P2 — an unsupervised latent that encodes the strategy

We train the VAE of Eq. (5) on occupancy histograms (Eq. 6) with no labels, and sweep the observation length L used to build each histogram. Recovery rises monotonically with L (Table 2, Figure 2): by $L = 100$ steps the IQL prey’s latent gives a supervised probe of 0.99 and an *unsupervised* ARI of 0.87, i.e. a two-component mixture on the latent recovers the placement almost perfectly. P2 holds for the IQL prey. For the MAPPO prey—more dominated even in the balanced regime—the latent retains strategy supervised (probe 0.83) but does not separate unsupervised (ARI ≈ 0); cleaner strategy expression yields a cleaner latent.

Table 2: **P2.** Placement recovery from the unsupervised VAE latent as a function of observation length L (IQL specialists). Occupancy-probe is the supervised upper bound on the histogram; latent-probe and ARI read the unsupervised latent.

	$L = 25$	$L = 50$	$L = 75$	$L = 100$
Occupancy-probe (supervised)	0.93	0.96	0.98	1.00
Latent-probe (supervised)	0.59	0.89	0.94	0.99
Latent ARI (unsupervised)	0.07	0.44	0.79	0.87

Past $L \approx 150$ the occupancy histogram saturates (the prey covers the whole arena) and the unsupervised ARI degrades even though the supervised latent-probe stays at 0.99; there is an intermediate observation window that is optimal for unsupervised strategy discovery. This is consistent with the planner needing a *sufficient but bounded* window of opponent observation to infer z .

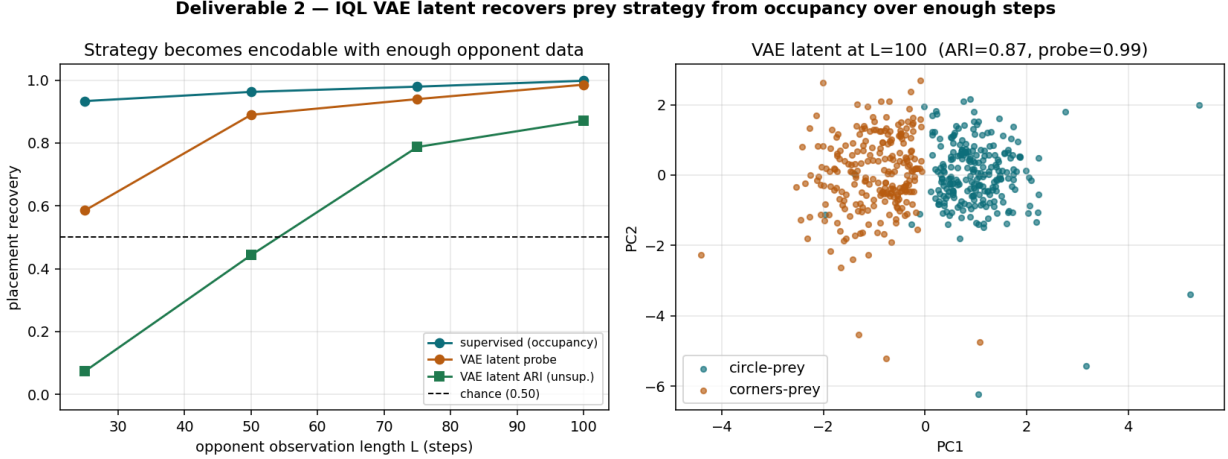


Figure 2: **P2.** Left: placement recovery vs. observation length; supervised occupancy-probe, supervised latent-probe, and unsupervised ARI all rise with L . Right: the unsupervised VAE latent at $L = 100$ forms two clusters aligned with circle vs. corners (ARI 0.87), trained without labels.

4.3 P3 — co-trained predators are brittle to an unseen opponent

Table 3 is the 2×2 cross-play matrix M of captures per episode (Eq. 3) for the MAPPO specialists; Figure 3 visualizes it. Reading column-wise holds the prey and its environment fixed and swaps only the predator, isolating the predator’s contribution: the circle-prey is caught 1.90 times per episode by its co-trained predator but only 1.11 by the unfamiliar one; the corners-prey, 1.29 vs. 0.72. The in-distribution mean is $\bar{M}_{\text{in}} = 1.59$ and the out-of-distribution mean is $\bar{M}_{\text{ood}} = 0.91$, a relative brittleness of $\beta = 0.43$. P3 holds for MAPPO. The IQL predators are too generic to specialize ($\bar{M}_{\text{in}} = 0.26$ vs. $\bar{M}_{\text{ood}} = 0.24$, $\beta = 0.07$), so brittleness is a property of the strong centralized-critic predator—precisely the policy class the opponent-aware planner is meant to improve.

Table 3: **P3.** Captures per episode (mean $\pm$ std over 3 seeds, 200 episodes each), MAPPO specialists. Rows: predator’s training layout; columns: prey’s training layout (each prey in its native environment). Diagonal = in-distribution, off-diagonal = out-of-distribution prey.

	prey: circle	prey: corners
pred: circle	1.90 $\pm$ 0.17	0.72 $\pm$ 0.08
pred: corners	1.11 $\pm$ 0.28	1.29 $\pm$ 0.37

5 Discussion

Regime dependence is the point, not a caveat. None of P1–P3 hold for the dominated prey of the prior experiment; all hold once the prey is allowed to express strategy. This is expected and, in fact, defines where opponent modeling is useful: an opponent that cannot vary its behavior has no strategy to infer, and a planner gains nothing by modeling it. The balanced, specialist regime is the operating point at which the proposal’s machinery is both necessary and applicable.

Deliverable 3 — co-trained predators are brittle to out-of-distribution opponents

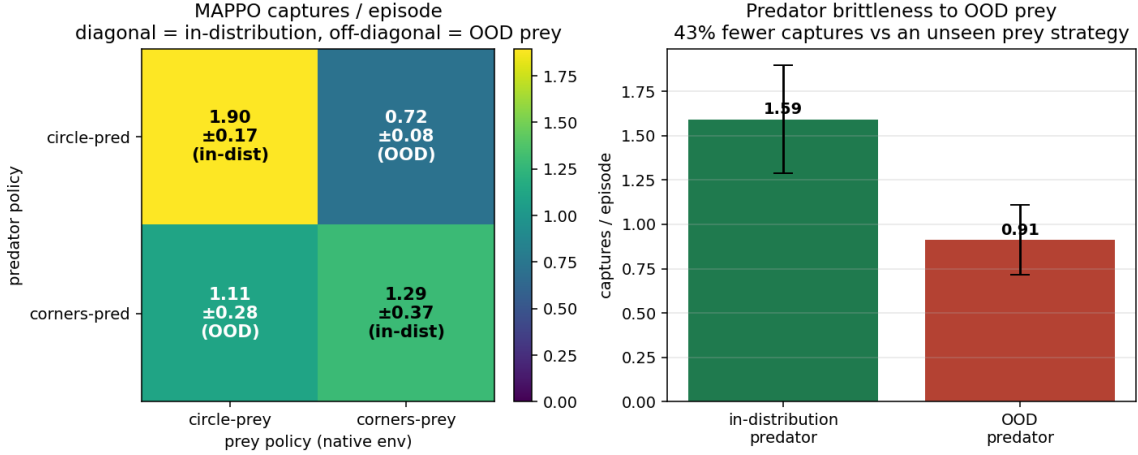


Figure 3: **P3.** Left: the cross-play matrix of captures/episode; the in-distribution diagonal is brighter than the OOD off-diagonal. Right: averaged over pairings, the co-trained predator catches 43% more than the unfamiliar one ($1.59 \rightarrow 0.91$).

A division of labor between algorithms. The cleanest *encodable* strategy comes from the value-based prey (P1, P2), whose weaker predators let it execute full routes; the cleanest *brittleness* comes from the MAPPO predator (P3), whose centralized critic learns a layout-specific interception that does not transfer. This is coherent: P1/P2 require red to express strategy, while P3 requires blue to be strong enough to overfit to it. A latent-conditioned planner is the bridge—it gives a decentralized executor access to the opponent structure that a centralized critic otherwise exploits only during training.

Representation and observation length. Two design choices were decisive for P2 and both transfer to the planned encoder. First, occupancy (Eq. 6) rather than the raw position sequence: the strategy is in *where* the opponent goes, and a reconstruction objective on ordered positions spends its capacity on evasion noise. Second, a sufficient observation window: separability is monotone in L up to saturation, so the planner’s belief over z should sharpen as red is observed and then be capped before the occupancy washes out.

6 Part 1 realized: a latent-conditioned opponent model

With P1–P3 established we build the proposal’s Part 1 directly: a single prey policy $\pi_\psi^{\text{red}}(a \mid s, z)$ that reproduces either specialist by conditioning on the latent. Let $z_\phi(\tau) = \mu_\phi(\Phi_B(\tau))$ be the encoder’s posterior mean (Eq. 5, the same $Z = 3$ occupancy VAE as P2). We log (s_t, a_t) pairs from the circle- and corners-specialist rollouts, tag each pair with its trajectory’s code, and minimize the behavior-cloning loss

$$\mathcal{L}_{\text{BC}}(\psi) = -\mathbb{E}_{\tau \sim \mathcal{D}} \mathbb{E}_{(s_t, a_t) \in \tau} \left[\log \pi_\psi^{\text{red}}(a_t \mid s_t, z_\phi(\tau)) \right]. \quad (7)$$

We then form class-mean codes $\bar{z}_\ell = \mathbb{E}_{\tau \in \ell} [z_\phi(\tau)]$ and probe *latent controllability*: roll π_ψ^{red} out with z *clamped* to $\bar{z}_{\text{circle}}$ vs. $\bar{z}_{\text{corners}}$ and measure whether the resulting routes differ. We report this

both across matched environments (each $\bar{z}_\ell$ in its native layout) and, more stringently, in a *fixed* environment with the same predators, where only z changes; the latter isolates the latent’s effect from the observation.

Table 4: **Part 1.** Latent-conditioned BC (IQL prey). Action-match is held-out accuracy of $\arg \max_a \pi_\psi(a \mid s, z_\phi(\tau))$ against the specialist’s action; route separation is the occupancy-probe accuracy separating the two clamped- z rollouts (chance 0.50).

Quantity	Value	Baseline / chance
BC action-match (held-out)	0.62	0.24 (majority action)
Encoder separates specialists	0.96	0.50
Route sep., matched env ($\bar{z}_\ell$)	1.00	0.50
Latent control, fixed env (swap z)	0.85	0.50

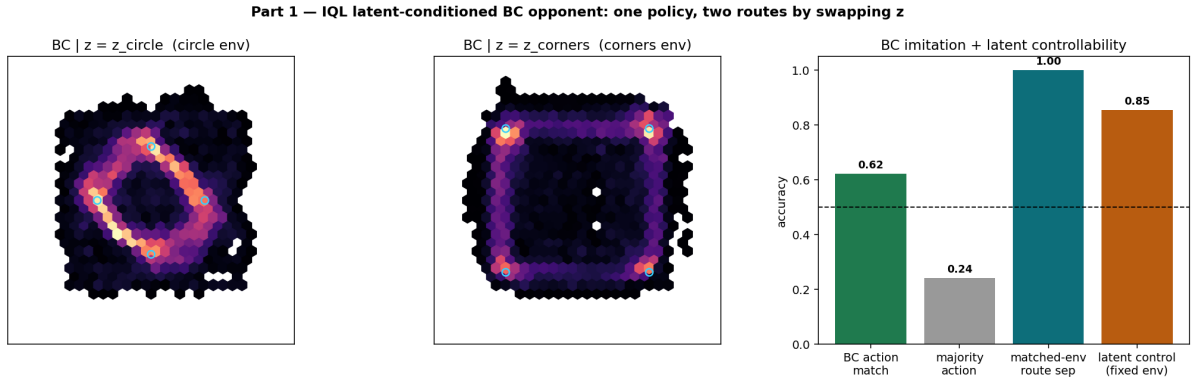


Figure 4: **Part 1.** One BC policy, two routes by swapping the latent. Left/middle: occupancy of π_ψ^{red} clamped to $\bar{z}_{\text{circle}}$ (a diamond) and $\bar{z}_{\text{corners}}$ (a square). Right: action-match exceeds the majority-action baseline, and the latent controls the route even with the environment held fixed (0.85).

The policy clones each specialist’s action at 0.62 held-out accuracy (vs. a 0.24 majority-action floor; the residual is the specialist’s own stochasticity given s), and—decisively for the planner—swapping z alone steers the prey’s route at 0.85 in a fixed environment (Table 4, Figure 4). This is the controllable opponent model Part 2 requires.

7 Outlook: Part 2

Part 2 embeds $\pi_\psi^{\text{red}}(\cdot \mid s, z)$ as the red-action sampler in a MuZero-style search and conditions the blue value function on z inferred from observed red play; the contrastive-preference variant replaces Eq. (7) with a Bradley–Terry objective whose negative trajectories are generated by the planner itself. P3 gives the concrete target, recovering on the OOD diagonal the captures a co-trained predator loses: the headline numbers to beat are in Table 3—an opponent-aware circle-predator should catch the corners-prey closer to the in-distribution 1.29 than the unaware 0.72.